

Course Description: Academic Year 2026-27

Course Title and Code: Programming - I (CS1139)	
Teaching Scheme and no. of hours per week	L-T-P: 3-0-2 (5 hours)
Credits	4
Programme(s) and Batch	BTech, 2026-30
Semester (as per curriculum structure)	I Sem
NCrF Level (4.5/5.0/5.5/6.0)	4.5
Pre-requisite course (Course Code and Title)	NA
Desired prior knowledge/skills	NA
Weightage	Theory 50%; Practical 50%
The occupational path it may lead to, if any	Python Developer; Python Engineer
Course Objective: This course aims to develop a clear understanding of the fundamentals of computer programming using Python. It covers the essential concepts required for students to write, execute, and understand programs independently. By the end of the course, students will be able to demonstrate problem-solving skills by designing and implementing algorithms to solve computational problems.	
Learning Outcomes: On successful completion of this course, the students should be able to: CS1139.1. Demonstrate problem-solving skills in Python by designing and implementing basic computing techniques and algorithms. CS1139.2. Apply appropriate Python data types and operators to develop solutions. CS1139.3. Design and implement programs using selection and iteration constructs. CS1139.4. Use lists, tuples, dictionaries, functions, and strings to solve problems. CS1139.5. Understand data/file handling techniques and exception handling.	

Syllabus

UNIT I: Basics of Computer and Programming: Flowcharts, Algorithms, Writing pseudocode, computational thinking, **Fundamentals of Python:** Beginning with Python, Parts of a Program: Modules, Statements and Expressions, Whitespace, Comments, Special Python Elements: Tokens, Naming Objects, Variables, Objects and Types, Operators, Relational Operators, Boolean Operators, Precedence.

UNIT II: Conditional Statements: The Selection Statement for Decisions: if, else if, and nested statements. **Repetition:** for statement, while statement, and nested loops.

UNIT III: Functions: What Is a Function? Python Functions, Flow of Control with Functions, Lambda Functions Scope, Arguments, Parameters, Default Values and Parameters, Functions as Objects; **Strings:** The String Type, String Operations, Formatted Output for Strings.

UNIT IV: Lists and Tuples: What Is a List? Iteration, Indexing and Slicing, Operators, Lists vs Strings, Split and Other Functions and Methods, Anagrams, Tuples from Lists. **Dictionaries and Sets:** Dictionaries, Python Dictionaries, Dictionary Indexing and Assignment, Sets, Python Sets, Methods, Operators, and Functions for Python Sets, Set Methods.

UNIT V: Files and Exceptions: *What Is a File? Accessing Files: Reading Text Files, Accessing Files: Writing Text Files, Reading and Writing Text Files in a Program, File Creation and Overwriting, Handling Errors: Error Names, the try-except Construct, try-except Flow of Control, Exception;*

Optional Topics: Libraries and Packages, Numpy, Pandas, Matplotlib, etc.

Text/Reference Books

1. David I. Schneider, 'An Introduction to Programming using Python' Pearson, 2016.
2. John V. Guttag, 'Introduction to Computation and Programming Using Python – with Applications to Computational Modeling and Understanding Data', MIT Press, 3rd Edition, 2021.
3. Introduction to Programming using Python – Y. Daniel Liang, Pearson.
4. William Punch, Richard Enbody, 'The Practice of Computing Using Python'. Pearson, 2016.
5. Allen B. Downey. Think Python. Green Tea Press, Massachusetts, USA.

Online Resources (Please provide link)

1. NPTEL: The Joy of Computing using Python
<https://nptel.ac.in/courses/106106182>
2. Infosys Springboard:
https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0135784773973196803862_shared/overview
3. Programming for Everybody (Getting started with Python)
<https://www.coursera.org/learn/python>
<https://www.coursera.org/specializations/python>
4. Python 3: Programming Specialization
<https://www.coursera.org/specializations/python-3-programming>

Evaluation Scheme:

S. No.	Specifications	Weightage	Marks
1	Class Participation	Nil	Nil
2	Assignment	Nil	Nil
3	Quiz	Nil	Nil
4	Mid-Term Examination (Theory)	20	20
5	End-Term Examination (Theory)	30	30
6	Report	Nil	Nil
7	Project (PBL)	20	20
8	Lab Evaluation	20	20
9	Course Portfolio	10	10
10	Presentation	Nil	Nil
11	Viva	Nil	Nil
	Total	100	100

Attendance Policy

Every student is expected to put in 100% attendance. However, the student with attendance below 75% in a course shall be penalised as follows.

- 70.0% to less than 75.0% attendance: A warning letter will be issued.

- 50.0% to less than 70.0% attendance: One grade drop. However, a D grade will remain D.
- Less than 50% attendance: Two grades drop. However, a D grade will remain D.

Note:

1. *This relaxation in attendance includes participation in all kinds of activities, medical emergencies, participation in placement opportunities, late registrations, disciplinary actions, or any other reasons not mentioned here.*
2. *Students will be responsible for maintaining attendance in the classes and monitoring their attendance online.*

Course Articulation Matrix: (Mapping of COs with POs)

Course Outcome	Correlation with program outcomes															Correlation with program specific outcomes	
	PO 1	PO 2a	PO 2b	PO 2c	PO 3a	PO 3b	PO 3c	PO 4a	PO 4b	PO 4c	PO 5a	PO 5b	PO 6	PO 7a	PO 7b	PSO 1	PSO 2
CS1139.1	2				2								1			2	1
CS1139.2	1					2	2	2	2			2	2		2	2	2
CS1139.3	2					2	2		1	2		1	1	1			1
CS1139.4	1				1		2		2			1		3		1	2
CS1139.5	1					1					2					2	

1- Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation

LESSON PLANNING (Theory)

Lecture	Unit/ Chapter/ Topic	Hours
Topic: Basics of Computer and Programming		4
1	Introduction to Computers	
2	Flowcharts	
3	Algorithms	
4	Writing pseudocode	
Topic: Fundamentals of Python		6
5	Beginning with Python, Parts of a Program: Modules	
6	Statements and Expressions, Whitespace, Comments	
7	Special Python Elements: Tokens, Naming Objects, Variables	
8	Objects and Types	
9	Operators, Relational Operators, Boolean Operators	
10	Precedence and Associativity	8
Topic: Conditional Statements and Repetition		
11	The Selection Statement for Decisions: if	
12	if-else, elif blocks and conditional logic	
13	for loops	
14	while loops	
15	break, continue, and pass statements	

16	Nested loops	
17	Infinite loops	
18	Patterns	
Topic: Functions and Strings		8
19	What Is a Function? Python Functions, Flow of Control	
20	Lambda Functions Scope, Arguments, Parameters	
21	Default Values and Parameters, Functions as Objects	
22	Recursion and functional concepts	
23	The String Type, String Operations	
24	Formatted Output for Strings (f-strings, format)	
25	String slicing and indexing	
26	Common string methods and manipulation	
Topic: Data Types in Python		8
27	What Is a List? Iteration, Indexing and Slicing, Operators	
28	Lists vs Strings, Split and Other Functions and Methods	
29	Anagrams, Tuples from Lists, Tuple operations	
30	Dictionaries, Python Dictionaries	
31	Dictionary Indexing and Assignment	
32	Sets, Python Sets	
33	Methods, Operators, and Functions for Python Sets	
34	Set Methods and comprehensive practice	
Topic: Files, Exceptions, and Advanced		2
35	What Is a File? Reading Text Files	
36	Writing Text Files in a Program, Overwriting	

LESSON PLANNING (Lab)

Lab	Laboratory
1	Setting up Python Environment (Spyder) and running basic scripts
2	Variables, Operators and expressions practice
3	Implementing conditional logic with if, elif, else
4	Writing for and while loops, generating patterns
5	Creating custom functions, working with parameters and return values
6	String manipulation and formatting exercises
7	List operations: appending, slicing, sorting, and comprehension
8	Working with Tuples and understanding immutability
9	Dictionary applications: counting frequencies, mapping keys to values
10	Set operations: union, intersection, difference in practical problems
11	File handling: reading from and writing to text files